Boehringer Ingelheim

IU Innovation Unit Research

A preliminary case study on smart biobank browsing and visualizing disease data

What is Innovation Unit?

Innovation unit is a research team at Boehringer Ingelheim dedicated in researching and documenting advanced biobank data such as variants, studies, or traits tied with genetic diseases.
The research here is about building and designing a search tool based on external and internal biobanks

My role

As the principle UX designer/researcher for this project, my role was about understanding the user requirements and the daily usage of searching disease records by lab experts online. Focus areas were mainly on filtering and data visualization.

Research

Wireframe

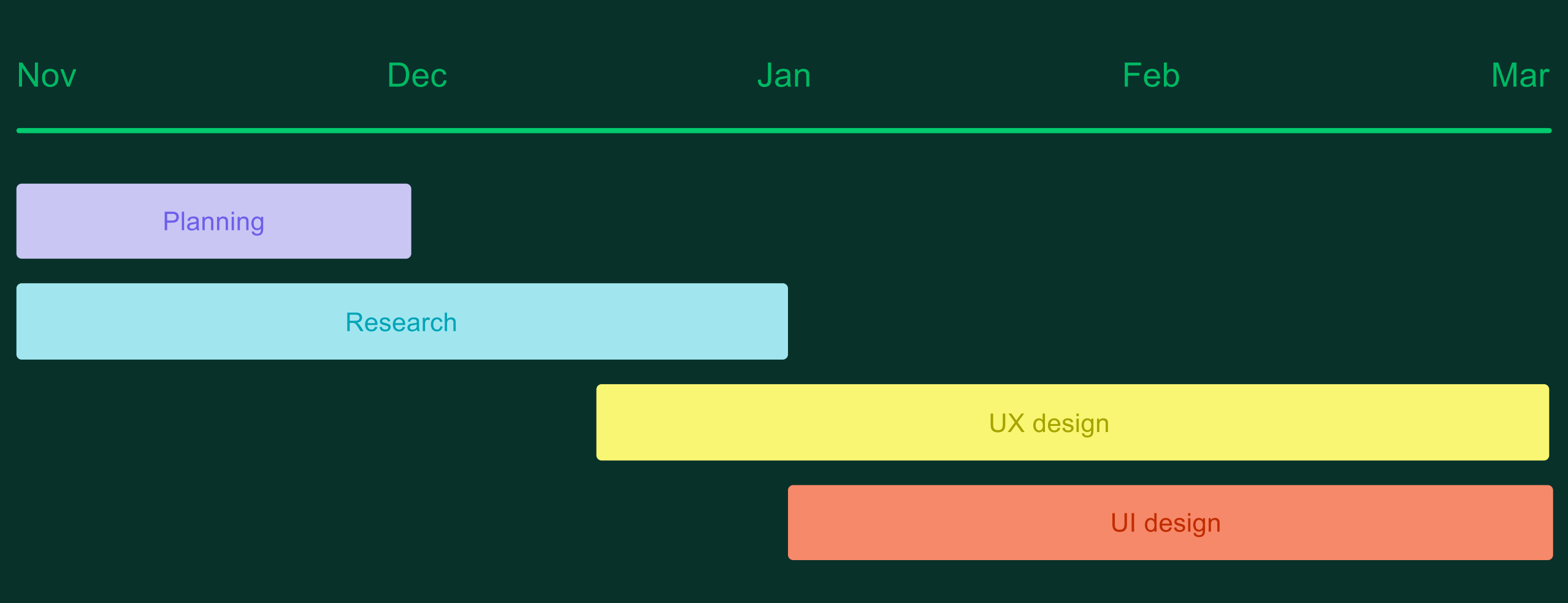
User personas

User flows

UI & UX design

Project timeline

As the timeline was limited to 9 months, the adequate amount of time had to be reserved for researching and interviewing experts



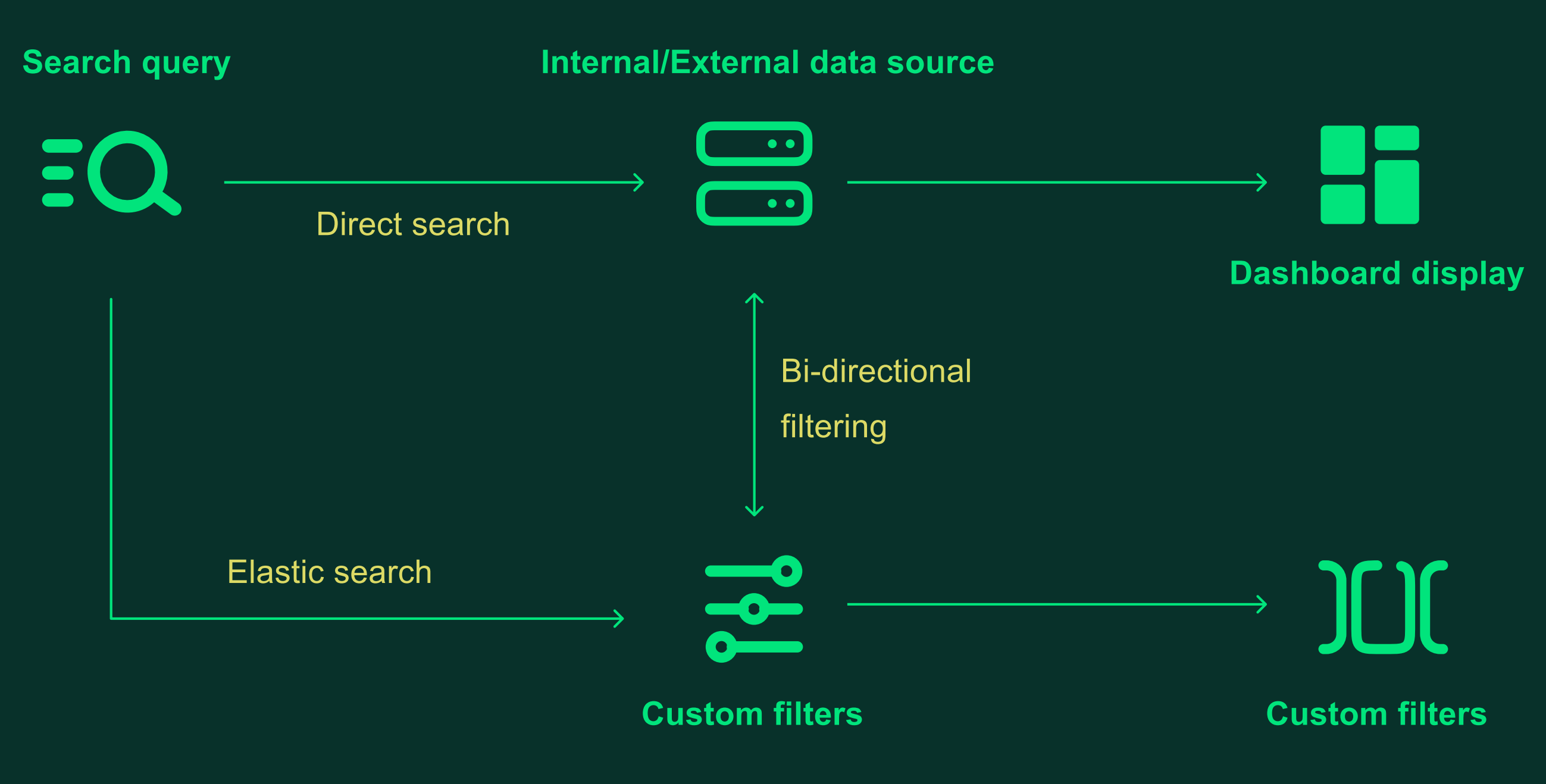
Requirement identification

Additional requirements such as multi-tier filtering and cross database searching were initially gathered during the process which lead to the idea of introducing a ***Smart search*** feature. The need for a user friendly, intuitive search and filter functionality was crucial as the targeted user base does not and will ***not*** spend more than 2-4 minutes in searching for a quick result.

Solution architecture

The solution was to retrieve data from multiple internal and external resources. The retrieval process will fetch all of the publicly accessible data of the searched disease and then output in text based and statistical forms. Few of the most prominent data attributes would be,

- Disease name
- Disease classification and type
- Associated diseases and variants
- Public figures (statistical) for each disease and it's associates



Displaying Quantitative outputs

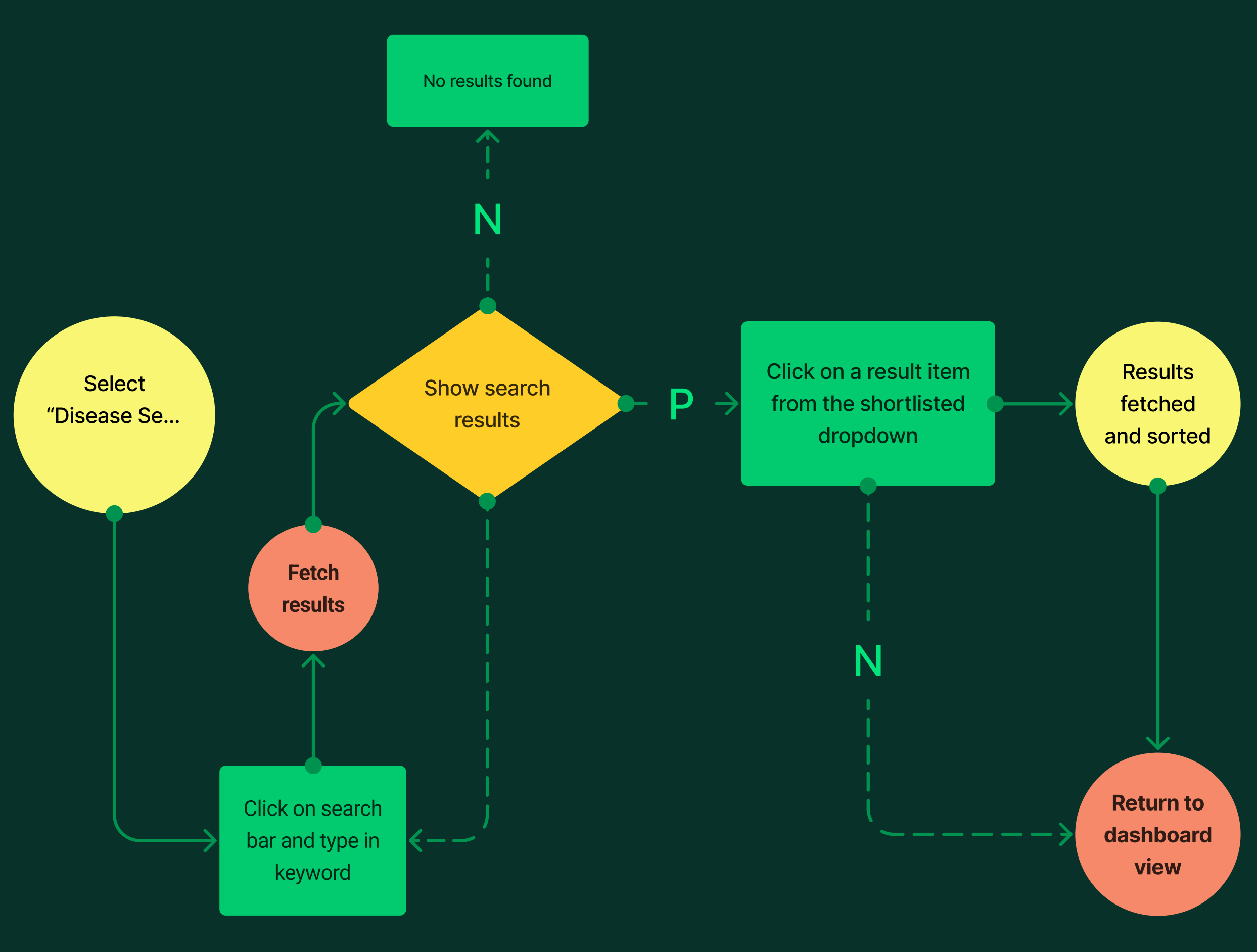
The sheer amount of data that will be output as numbers had to be categorized and divided into subsections. As Elastic search triggers data from multiple domains, the categorizing had to be based on the source of the data.
Objective approach in solving the issue likely prompted showing straight forward data instead of getting biased/hypothetical outputs.

Example

✓ Conversion of input genotype data to PLINK2 pgen/pvar/psam format and subsetting to only include individuals present in the input phenotype data.

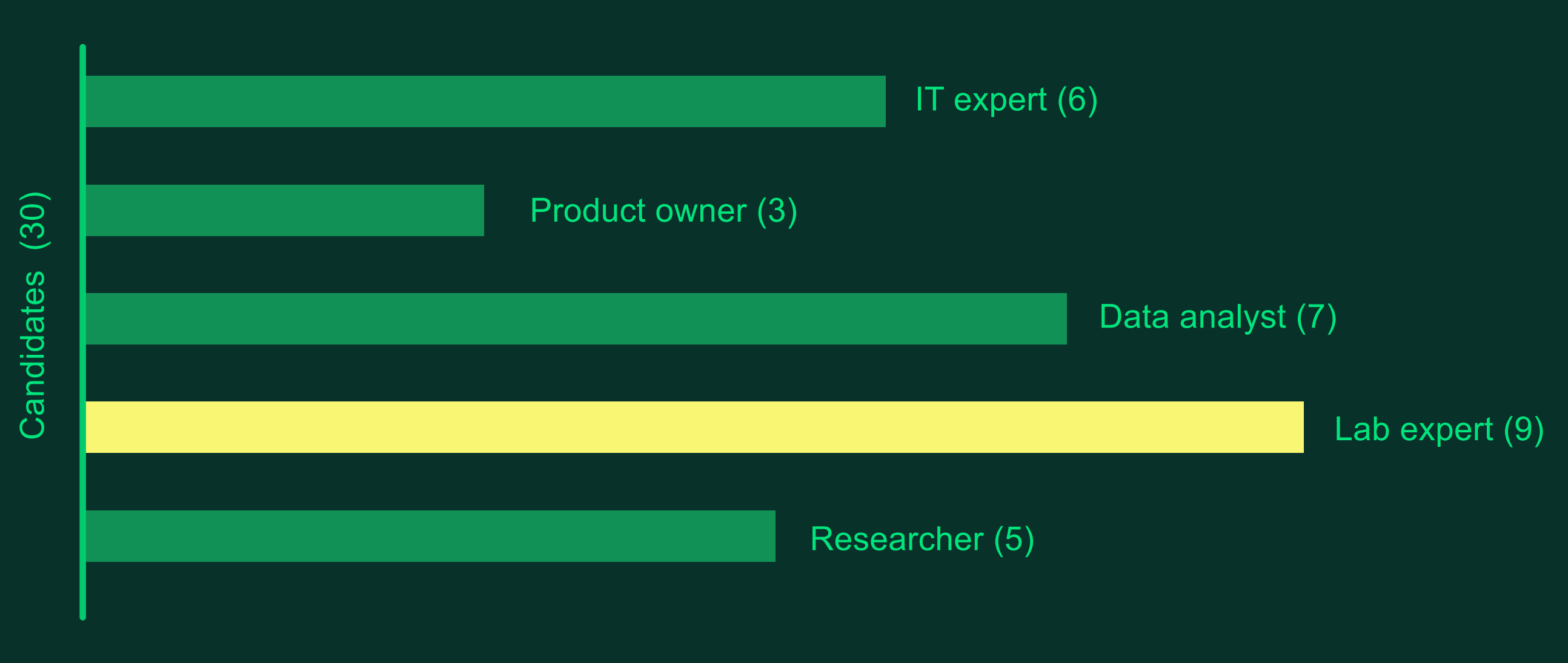
Input genotype data to PLINK2 pgen/pvar/psam format and subsetting and ***might*** only include individuals present in the input phenotype data.

Below shows the ideal workflow for the results filtering component


No results found
N
Select "Disease Se...
Show search results
P
Click on a result item from the shortlisted dropdown
Results fetched and sorted
Fetch results
Click on search bar and type in keyword
N
Return to dashboard view

User base and preferences

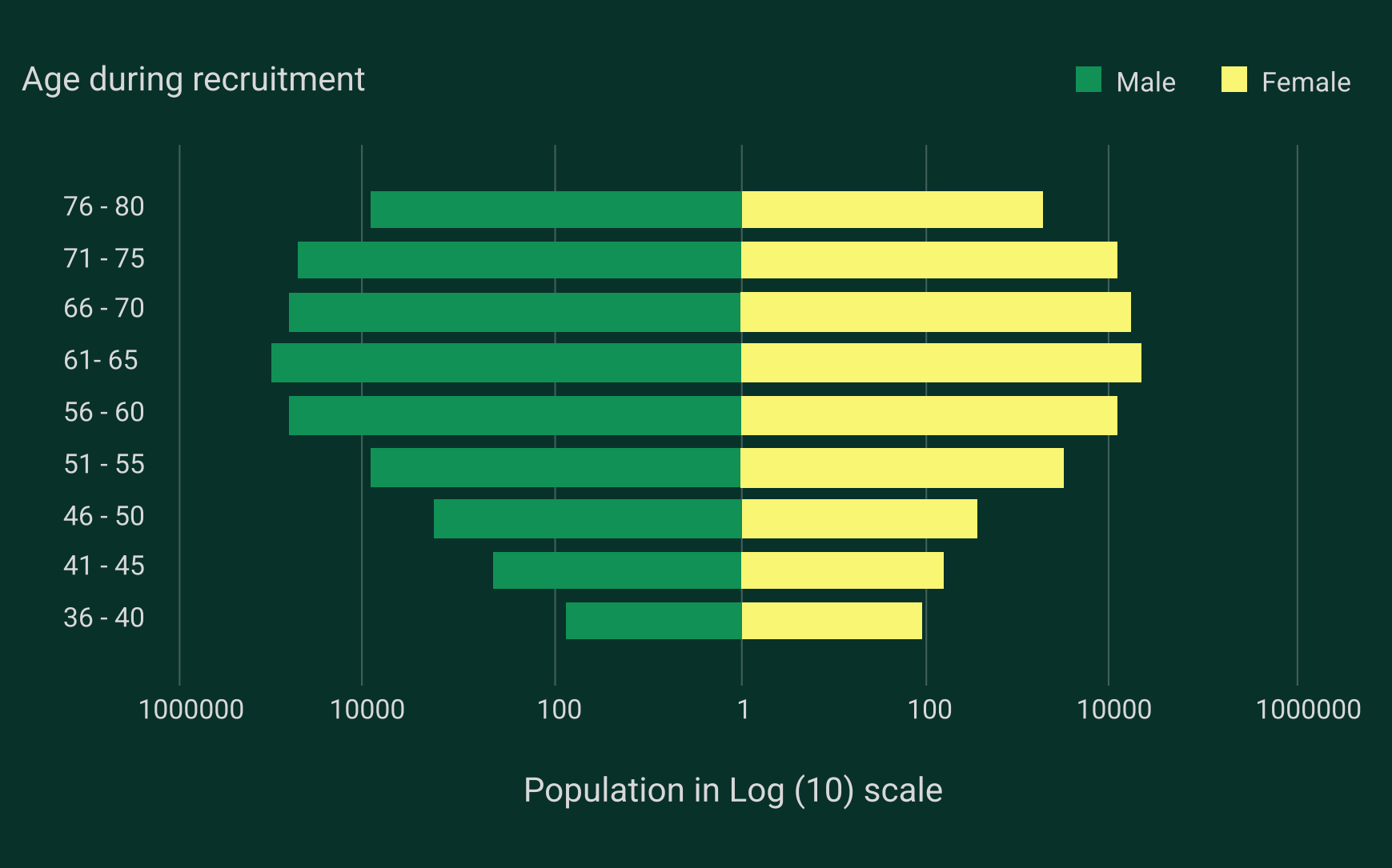
The initial user research was held among **30 handpicked** candidates with diverse user roles and are potential users of the application. he below graph shows



This quantitative analysis with the diverse amount of candidates allowed to have an in-depth understanding of how the tool will be used and how the information is read. The feedback from both technical experts and SME's (Lab experts, researchers) was undeniably important as the feedback received for data readability and visualization aspects were vary and the solution had to be fine tuned considering both feedbacks to close the gap between the two user groups.

Visualization of data

Visualizing data was a key element of the project, as it assures readability, accuracy and is compliant with the main user group's expertise and understanding. In terms of usability, the charts had to be hand picked and clarified with the user groups beforehand in creating the prototypes.

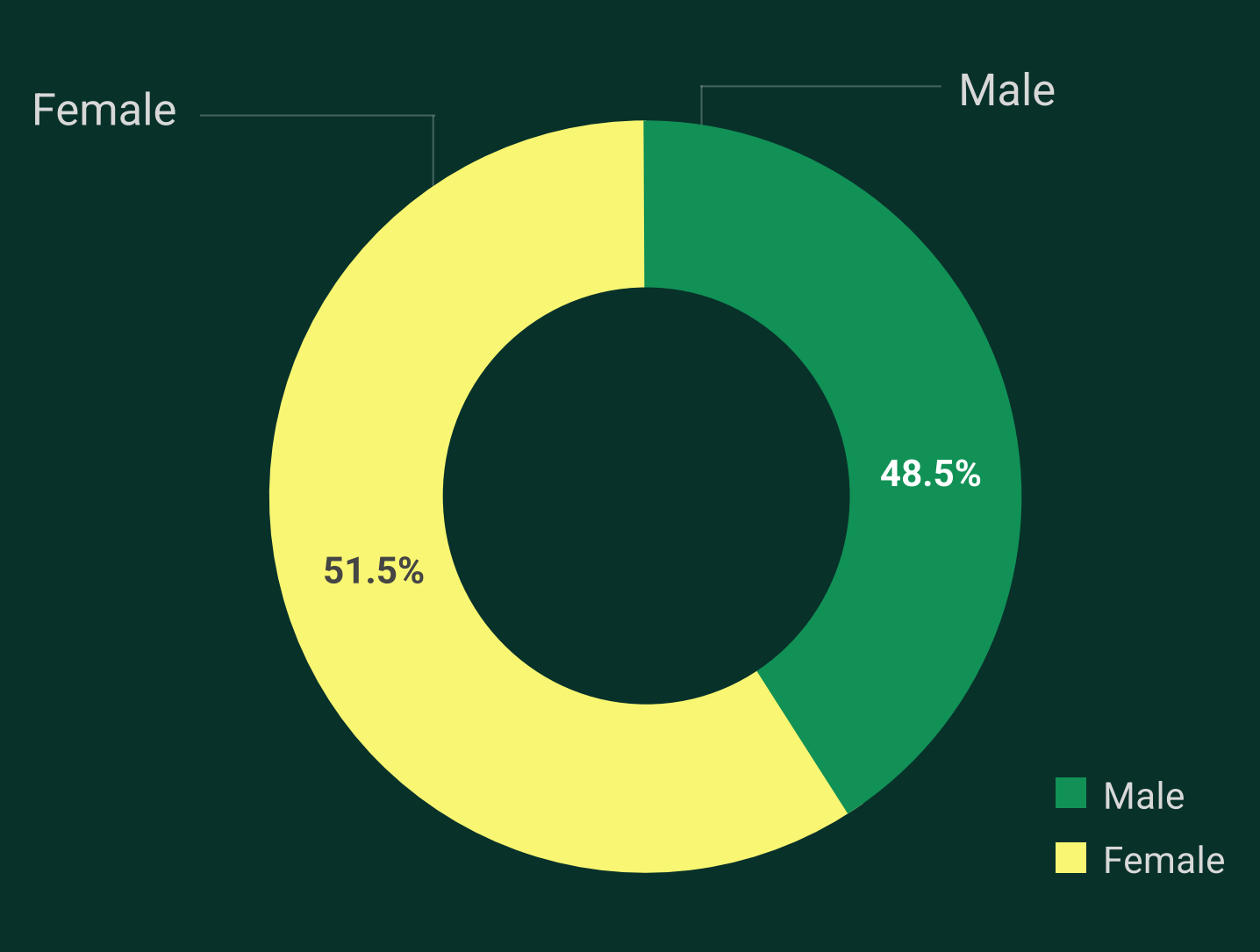


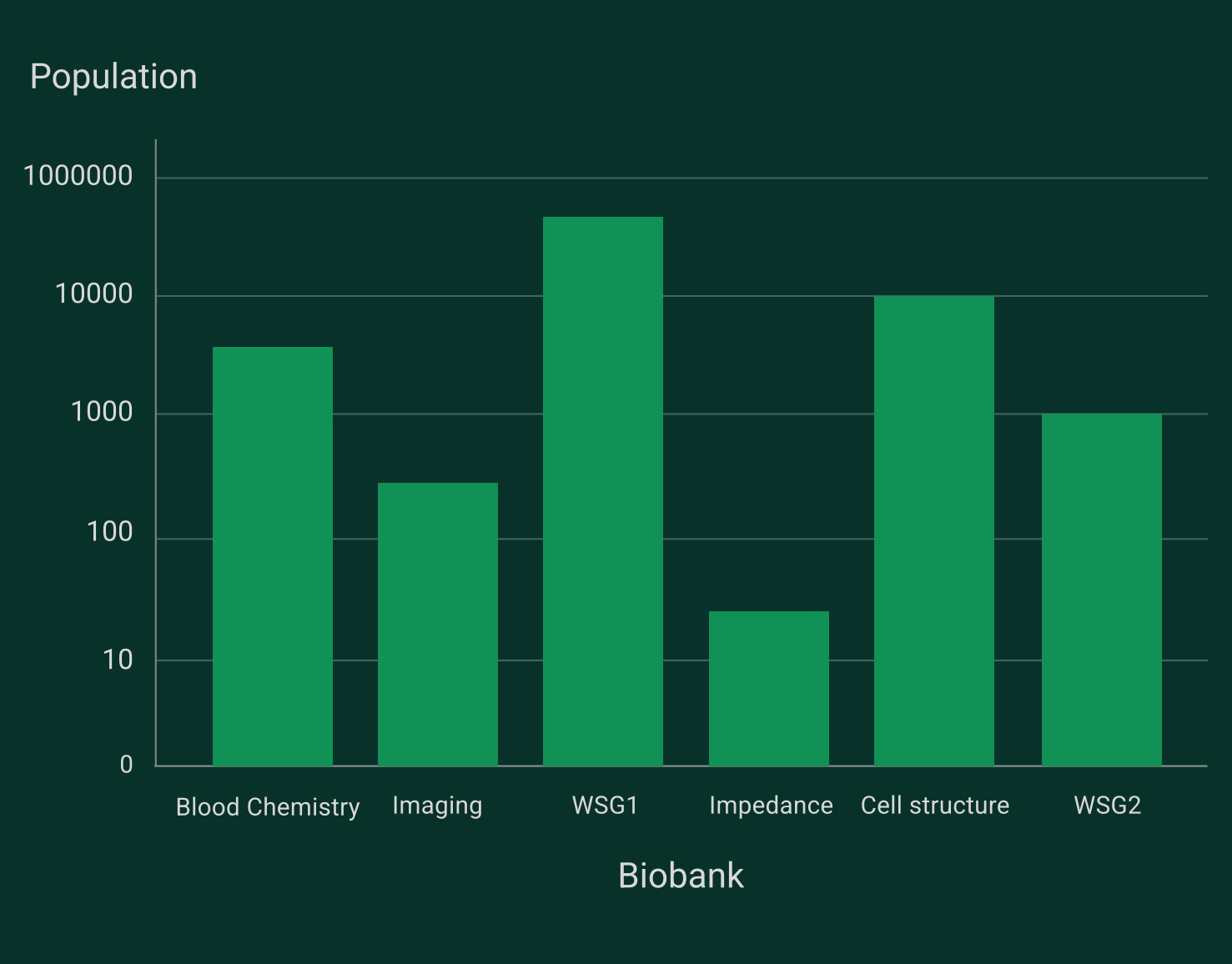
Bi-directional bar chart

Componential Bi-directional bar chart is being used here to visualize and compare the data for both gender variations in a log(10) scale.

Pie/Donut chart

A simple Pie chart with the zoom-in ability was also introduced to have an alternative perspective of the same data types/sets. Percentage based will switch to numerical values when on hover.



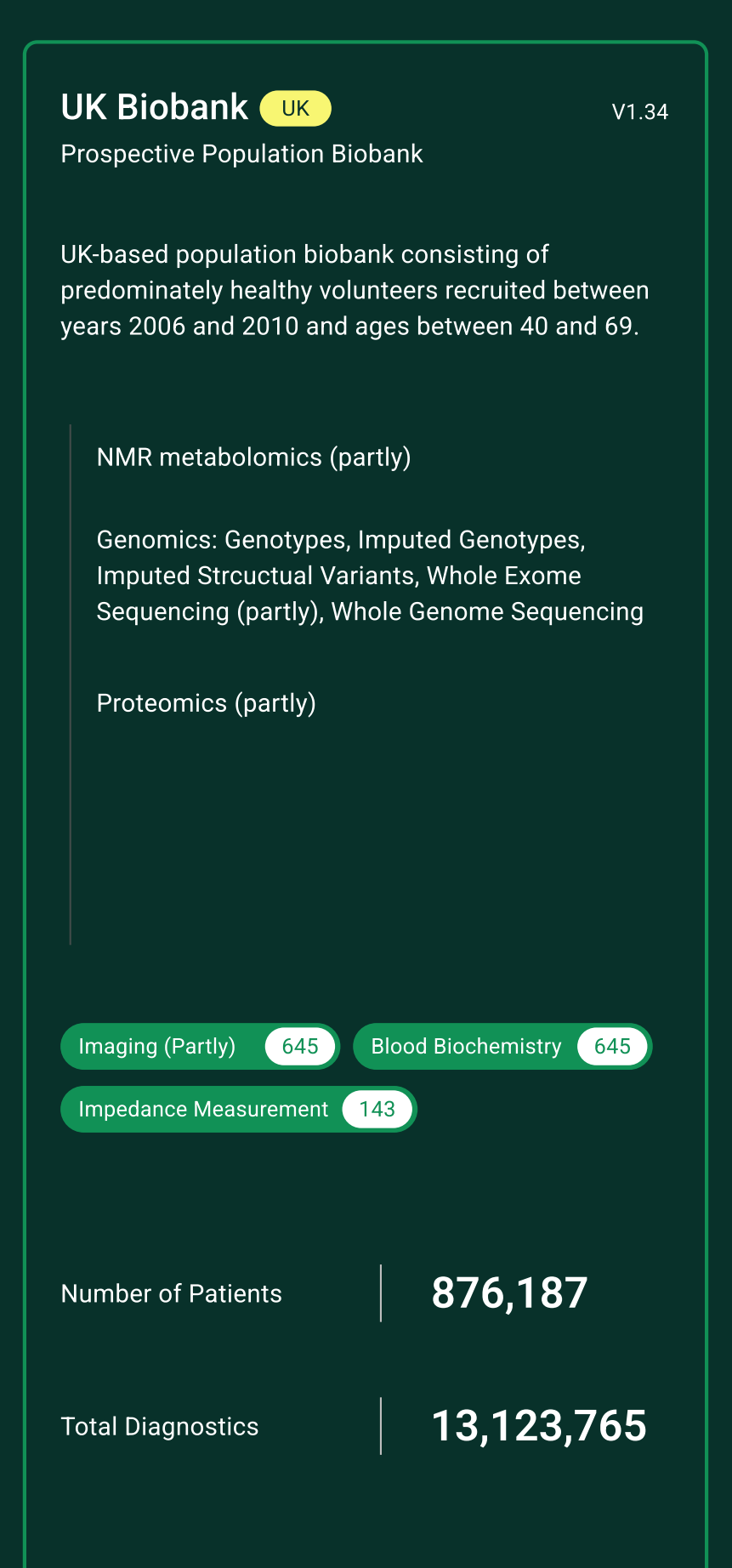


Pie/Donut chart

Horizontal bar chart was used to show the categorization between different collection scopes within Biobanks.

Furthermore, as D3.js library allowed fully customizability, spacing and text sizes were also adjusted to suit the over the design and branding guidelines.

Rendering Biobank results



UK Biobank UK V1.34

Prospective Population Biobank

UK-based population biobank consisting of predominately healthy volunteers recruited between years 2006 and 2010 and ages between 40 and 69.

NMR metabolomics (partly)

Genomics: Genotypes, Imputed Genotypes, Imputed Strcuctual Variants, Whole Exome Sequencing (partly), Whole Genome Sequencing

Proteomics (partly)

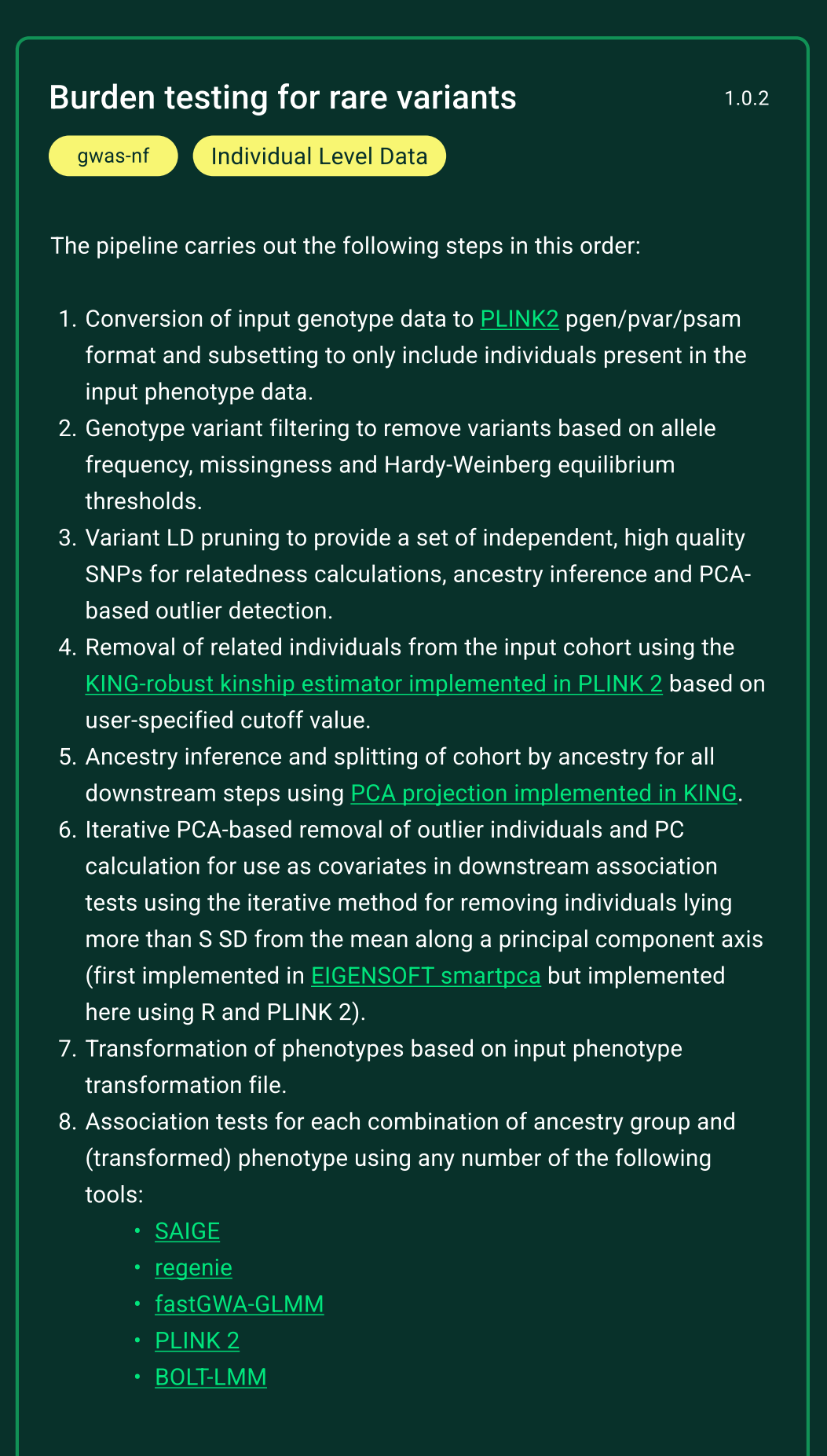
Imaging (Partly) 645 | Blood Biochemistry 645 | Impedance Measurement 143

Number of Patients	876,187
Total Diagnostics	13,123,765

The final rendering of the information which is extracted from the respective Biobanks are shown in a dedicated sidebar for easy viewing. This section also included,

- External links
- Biobank classifications*
- Patient demographies

for comparison reasons



Burden testing for rare variants 1.0.2

gwas-nf | Individual Level Data

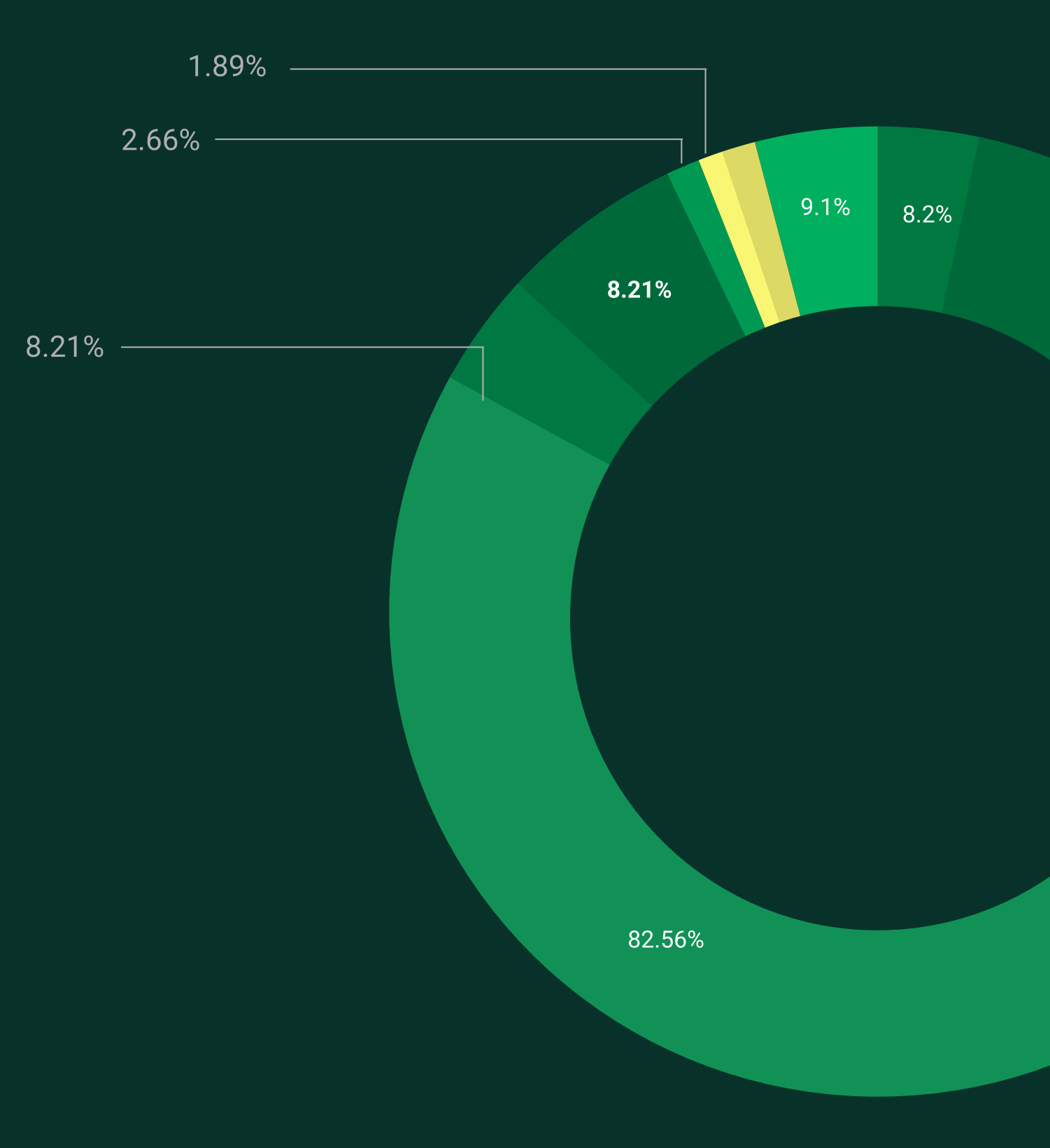
The pipeline carries out the following steps in this order:

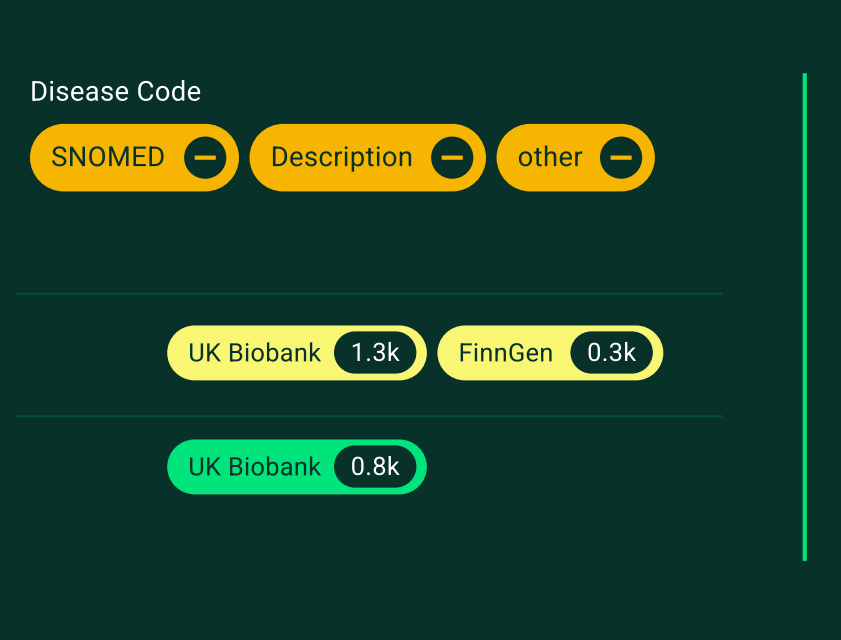
1. Conversion of input genotype data to PLINK2 pgen/pvar/psam format and subsetting to only include individuals present in the input phenotype data.
2. Genotype variant filtering to remove variants based on allele frequency, missingness and Hardy-Weinberg equilibrium thresholds.
3. Variant LD pruning to provide a set of independent, high quality SNPs for relatedness calculations, ancestry inference and PCA-based outlier detection.
4. Removal of related individuals from the input cohort using the KING-robust kinship estimator implemented in PLINK 2 based on user-specified cutoff value.
5. Ancestry inference and splitting of cohort by ancestry for all downstream steps using PCA projection implemented in KING.
6. Iterative PCA-based removal of outlier individuals and PC calculation for use as covariates in downstream association tests using the iterative method for removing individuals lying more than S SD from the mean along a principal component axis (first implemented in EIGENSOFT smartpca but implemented here using R and PLINK 2).
7. Transformation of phenotypes based on input phenotype transformation file.
8. Association tests for each combination of ancestry group and (transformed) phenotype using any number of the following tools:
    - SAIGE
    - regenie
    - fastGWA-GLMM
    - PLINK 2
    - BOLT-LMM

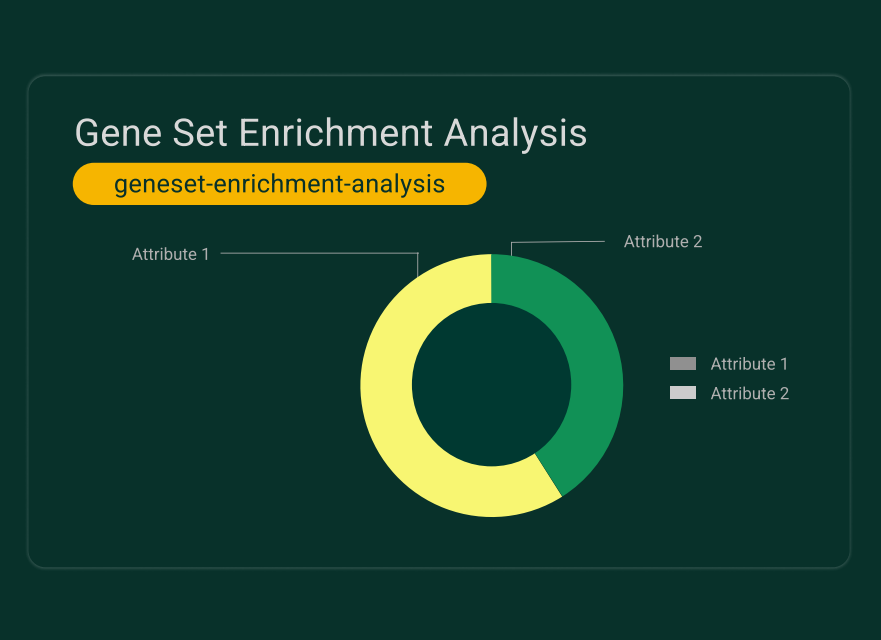
Report generation

Ultimately, based on the findings and search criteria via the smart filtering tool, a downloadable PDF document will be generated with the rendered graphical charts for further offline assessments.

This feature us particularly interested within the lab experts as the report will be documented physically for later references and assessment.







Color coding and prioritization

Colors throughout the application is formed in a way to highlight the pint of interest based on values, prioritization of information and result accuracy.
In order to achieve this,
Orange 🟧,Faded yellow 🟨,Light green 🟩 has been used in alignment with the new branding color guidelines at Boehringer Ingelheim

Project details

18 Weeks

5 Screens

8 Team members

Technology/Tool stack